TEC | Tecnológico de Costa Rica

Computer Engineering Department
Computer Engineering
IC6400 - Operations Research
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Simplex Method Project

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George Dantzig

- George Bernard Dantzig (1914-2005) was an American mathematician and computer scientist.
- He is best known for his development of the Simplex algorithm, a method for solving linear programming problems.
- Dantzig's work in optimization and operations research has had a profound impact on various fields, including economics, engineering, and logistics.
- He made significant contributions to the theory of linear programming, integer programming, and network flow problems.
- Dantzig's work has been widely recognized and has earned him numerous awards and honors throughout his career.

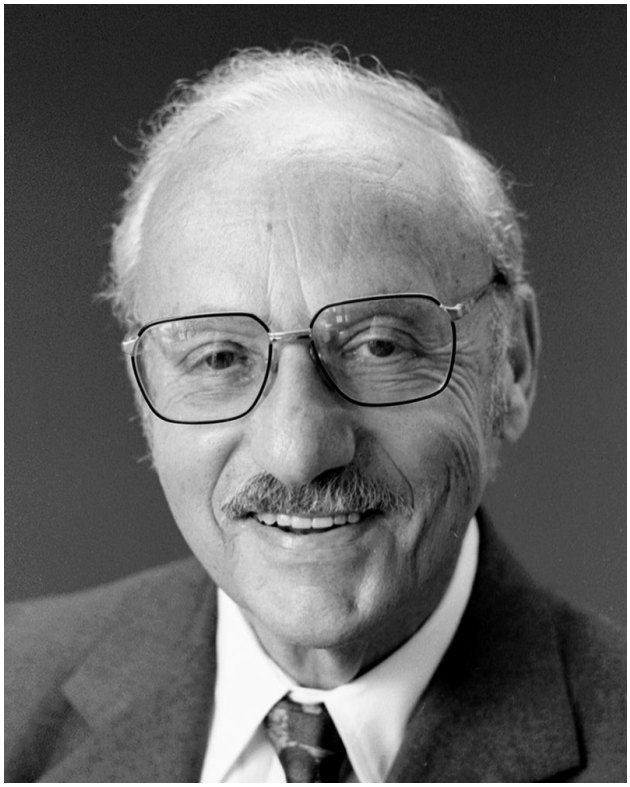


Figure 1: Image inspired by George Dantzig, the creator of the Simplex Algorithm.

The Simplex Algorithm

- It is a "mechanical" method that allows to find optimal solutions to linear programming problems.
- It was invented by George Dantzig in 1947.
- Uses matrix operations until it finds the optimal solution or determines that the problem is unbounded or infeasible.
- It is based in a geometric interpretation of the problem, where the feasible region is a convex polytope and the optimal solution is found at one of its vertices.

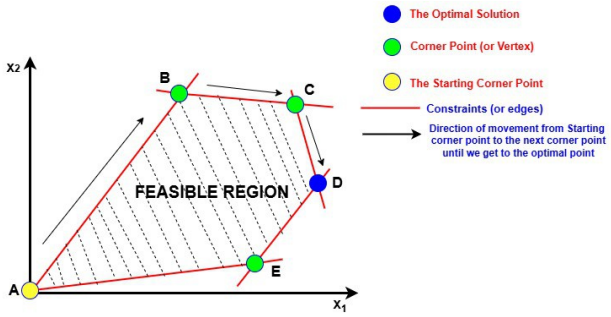


Figure 2: Image inspired by the geometric interpretation of the Simplex Algorithm.

Original Problem

Description

- Problem Name: Problema 7
- Amount of decision variables: 2
- Amount of restrictions: 2
- Total Amount of variables (decision + slack): 4
- Type of problem: Maximization

Equations

$$
\begin{aligned}
&\text{Max } Z = 2\,x_2 \\
&\text{subject to:} \\
&\qquad x_1 - x_2 \leq 4 \\
&\qquad -x_1 + x_2 \leq 1 \\
&\qquad x_1, x_2 \geq 0
\end{aligned}
$$

Initial Table

Z	x_1	x_2	s_1	s_2	b
1.0000	0.0000	-2.0000	0.0000	0.0000	0.0000
0.0000	1.0000	-1.0000	1.0000	0.0000	4.0000
0.0000	-1.0000	1.0000	0.0000	1.0000	1.0000

Intermediate Tables

Iteration 1

Entering variable: x_2 (most negative coefficient in the Z row: -2.0000). Leaving variable: s_2 (row 2, minimum ratio 1.0000).

Finding the Pivot

Z	x_1	x_2	s_1	s_2	b	P
1.0000	0.0000	-2.0000	0.0000	0.0000	0.0000	-
0.0000	1.0000	-1.0000	1.0000	0.0000	4.0000	-
0.0000	-1.0000	**1.0000**	0.0000	1.0000	1.0000	1.0000

Canonize the Pivot column with Pivot value

Z	x_1	x_2	s_1	s_2	b
1.0000	-2.0000	0.0000	0.0000	2.0000	2.0000
0.0000	0.0000	0.0000	1.0000	1.0000	5.0000
0.0000	-1.0000	1.0000	0.0000	1.0000	1.0000

Iteration 2

The entering variable would be x_1 (coefficient -2.0000 in the Z row), but every entry of its column in the constraints is less than or equal to zero, so no row limits its growth: the problem is **unbounded**.

Z	x_1	x_2	s_1	s_2	b
1.0000	-2.0000	0.0000	0.0000	2.0000	2.0000
0.0000	0.0000	0.0000	1.0000	1.0000	5.0000
0.0000	-1.0000	1.0000	0.0000	1.0000	1.0000

Final Table

Z	x_1	x_2	s_1	s_2	b
1.0000	-2.0000	0.0000	0.0000	2.0000	2.0000
0.0000	0.0000	0.0000	1.0000	1.0000	5.0000
0.0000	-1.0000	1.0000	0.0000	1.0000	1.0000

Solution

Solution

The problem is **unbounded**: there is no finite optimal solution.

Unbounded problem

In iteration 2 the variable x_1 has an improving coefficient in the Z row (-2.0000), but all the entries of its column in the constraint rows are ≤ 0. Therefore the ratio test has no candidate row and x_1 can grow indefinitely without violating any constraint, improving Z without limit. The problem has no finite optimal solution.